A Canonical Pretree Attached to a Cotlar Symbol

Automatic math-discovery packet

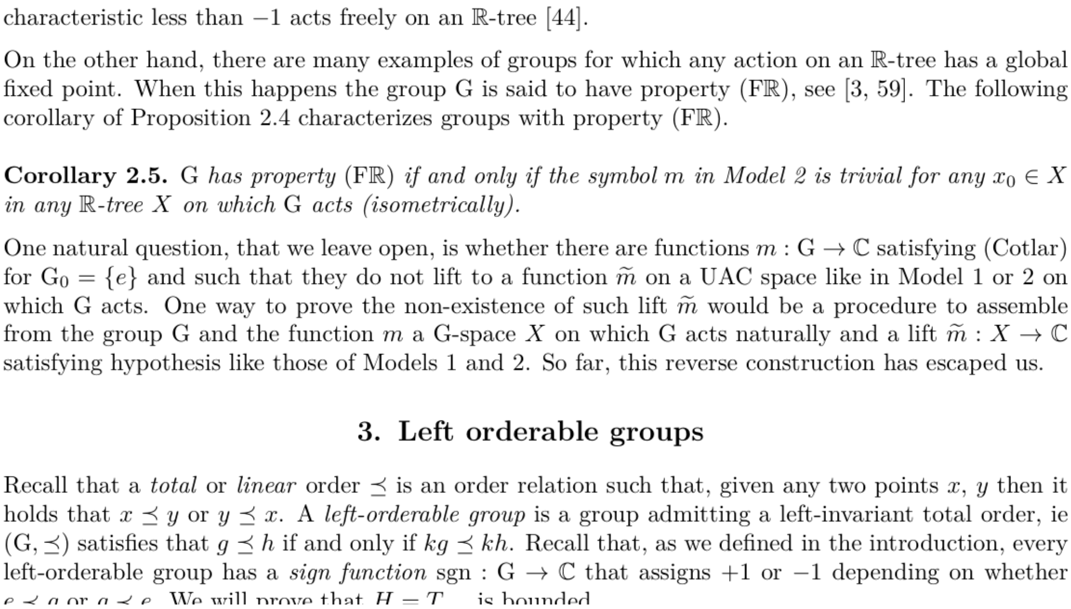
July 3, 2026

Source question

The source paper is Adrian M. Gonzalez-Perez, Javier Parcet, and Runlian Xia, *Noncommutative Cotlar identities for groups acting on tree-like structures*, arXiv:2209.05298. In the discrete case with $G_0 = \{e\}$, its closed formula says that a Fourier multiplier symbol $m : G \to \mathbb{C}$ satisfies the noncommutative Cotlar identity precisely when

$$\big(m(g^{-1}) - m(h)\big)\big(m(gh) - m(g)\big) = 0, \qquad g \neq e,\ h \in G. \tag{1}$$

After constructing examples from actions on uniquely arcwise connected spaces, the source asks whether every symbol satisfying Cotlar lifts to such a UAC model, or whether there are symbols outside Models 1 and 2.



characteristic less than -1 acts freely on an $\mathbb{R}$-tree [44].

On the other hand, there are many examples of groups for which any action on an $\mathbb{R}$-tree has a global fixed point. When this happens the group G is said to have property (F$\mathbb{R}$), see [3, 59]. The following corollary of Proposition 2.4 characterizes groups with property (F$\mathbb{R}$).

Corollary 2.5. G *has property* (F$\mathbb{R}$) *if and only if the symbol* m *in Model 2 is trivial for any* $x_0 \in X$ *in any* $\mathbb{R}$*-tree* X *on which* G *acts (isometrically).*

One natural question, that we leave open, is whether there are functions $m : \mathrm{G} \to \mathbb{C}$ satisfying (Cotlar) for $\mathrm{G}_0 = \{e\}$ and such that they do not lift to a function $\widetilde{m}$ on a UAC space like in Model 1 or 2 on which G acts. One way to prove the non-existence of such lift $\widetilde{m}$ would be a procedure to assemble from the group G and the function m a G-space X on which G acts naturally and a lift $\widetilde{m} : X \to \mathbb{C}$ satisfying hypothesis like those of Models 1 and 2. So far, this reverse construction has escaped us.

3. Left orderable groups

Recall that a *total* or *linear* order $\preceq$ is an order relation such that, given any two points x, y then it holds that $x \preceq y$ or $y \preceq x$. A *left-orderable group* is a group admitting a left-invariant total order, ie $(\mathrm{G}, \preceq)$ satisfies that $g \preceq h$ if and only if $kg \preceq kh$. Recall that, as we defined in the introduction, every left-orderable group has a *sign function* $\mathrm{sgn} : \mathrm{G} \to \mathbb{C}$ that assigns $+1$ or -1 depending on whether $e \prec g$ or $g \prec e$. We will prove that $H = T$ is bounded

Figure 1: The source's reverse-UAC question, on page 21 of the arXiv PDF.

Claim

This packet proves that the algebraic part of the reverse construction is automatic. Every m satisfying (1) canonically defines a G-invariant pretree structure on the orbit G. At the root e, the pretree shadows are exactly the level sets of m. Thus the remaining gap to a full positive answer is a topological realization problem for this pretree.

Proof intuition

In a tree, removing a point b partitions the other points into components. For an orbit $G \cdot x_0$, the component of a after removing b is read by translating b to the root and looking at $b^{-1}a$. This suggests declaring b to be between a and c exactly when

$$m(b^{-1}a) \neq m(b^{-1}c).$$

The Cotlar formula is then precisely the no-crossing rule for tree betweenness: if b separates a from c, then c cannot also separate a from b. The remaining pretree axiom is just the pigeonhole fact that if two colors differ, any third color differs from at least one of them.

Pretree theorem

We use the standard pretree axioms: a ternary relation $\langle a, b, c\rangle$ satisfies

$$\begin{array}{ll} \text{(B1)} & \langle a,b,c\rangle \Rightarrow \langle c,b,a\rangle, \\ \text{(B2)} & \langle a,b,c\rangle \ \&\ \langle a,c,b\rangle \iff b = c, \\ \text{(B3)} & \langle a,b,c\rangle \Rightarrow \langle a,b,d\rangle \text{ or } \langle d,b,c\rangle. \end{array}$$

These are the axioms used, for example, in the pretree and shadow-topology literature of Bowditch and Malyutin.

Theorem 1. *Let G be a group and let $m : G \to \mathbb{C}$ satisfy* (1)*. Define a ternary relation R_m on G by declaring $\langle a, b, c\rangle_m$ if either $b = a$, or $b = c$, or a, b, c are pairwise distinct and*

$$m(b^{-1}a) \neq m(b^{-1}c).$$

Then (G, R_m) is a pretree. Moreover, the left action of G on itself preserves R_m.

Proof. The relation is symmetric in the outer variables, so (B1) is immediate.

We prove (B2). If $b = c$, then both $\langle a, b, c\rangle_m$ and $\langle a, c, b\rangle_m$ hold by the endpoint convention. Conversely, suppose $b \neq c$ and $\langle a, b, c\rangle_m$. If one of the entries is an endpoint case, the assertion is immediate, so assume a, b, c are pairwise distinct. Then

$$m(b^{-1}a) \neq m(b^{-1}c).$$

Set

$$g = c^{-1}b, \qquad h = b^{-1}a.$$

Then $g \neq e$, $g^{-1} = b^{-1}c$, and $gh = c^{-1}a$. Since

$$m(g^{-1}) = m(b^{-1}c) \neq m(b^{-1}a) = m(h),$$

the first factor in (1) is nonzero. Therefore the second factor must vanish:

$$m(c^{-1}a) = m(c^{-1}b).$$

Thus $\langle a, c, b\rangle_m$ fails. This proves (B2).

It remains to prove (B3). Assume $\langle a, b, c\rangle_m$. Endpoint cases are again immediate, so suppose a, b, c are pairwise distinct and

$$m(b^{-1}a) \neq m(b^{-1}c).$$

Let $d \in G$. If $d = b$, then both alternatives are endpoint cases. If

$$m(b^{-1}a) \neq m(b^{-1}d),$$

then $\langle a, b, d\rangle_m$. Otherwise

$$m(b^{-1}d) = m(b^{-1}a),$$

and hence $m(b^{-1}d) \neq m(b^{-1}c)$, so $\langle d, b, c\rangle_m$. This proves (B3).

Finally, for $k \in G$,

$$(kb)^{-1}(ka) = b^{-1}a, \qquad (kb)^{-1}(kc) = b^{-1}c.$$

Therefore $\langle a, b, c\rangle_m$ iff $\langle ka, kb, kc\rangle_m$. The left action preserves the pretree relation. □

Root shadows

Corollary 1. *For $g, h \in G \setminus \{e\}$,*

$$e \text{ lies between } g \text{ and } h \iff m(g) \neq m(h).$$

Thus the equivalence classes of $G \setminus \{e\}$ not separated by the root are exactly the nonidentity level sets of m.

Proof. This is the definition of R_m with $b = e$:

$$m(e^{-1}g) \neq m(e^{-1}h) \iff m(g) \neq m(h).$$

□

Conditional lift

If the pretree (G, R_m) admits an equivariant realization inside a UAC space X, with G acting by homeomorphisms and with betweenness on the orbit agreeing with R_m, then the source's reverse construction follows for m. Indeed, take $x_0 = e$ under the orbit identification. By Corollary 1, the connected components of $X \setminus \{x_0\}$ that meet the orbit are labeled by the values of m. Define $\widetilde{m}(x_0) = m(e)$ and make $\widetilde{m}$ constant with the corresponding value on each such component, assigning arbitrary bounded values to any component that misses the orbit. Then

$$\widetilde{m}(g \cdot x_0) = m(g) \qquad (g \in G),$$

and $\widetilde{m}$ is constant along arcs in $X \setminus \{x_0\}$. Since the left action on the orbit has stabilizer $\{e\}$, this is exactly the Model 1 pattern for $G_0 = \{e\}$.

Scope and limitations

This packet is not a full solution of the reverse-UAC problem. The proved part is the canonical algebraic pretree construction. The remaining step is topological: prove, in the required generality, that this pretree has an equivariant realization as a uniquely arcwise connected topological space whose root components realize the pretree shadows. Standard pretree, shadow-topology, dendron, and order-tree realization results suggest this is the correct route, but this packet does not claim that realization theorem.

Relation to the torsion packet

The earlier packet

`solutions/partial/2209.05298_torsion_cotlar_collapse/`

proved that on torsion groups, Cotlar symbols are constant off the identity. The present theorem explains the broader structure behind that collapse: a Cotlar symbol is not an arbitrary coloring of the group, but a coloring of the root shadows of a canonical pretree.

Novelty and search status

The direct searches used for the torsion packet were repeated conceptually for this stronger structural route, including phrases around the source title, "reverse construction", "UAC", "Cotlar torsion", and "pretree Cotlar". The source paper and general pretree/dendron references appeared, but no paper was found stating this Cotlar-to-pretree construction as an answer to the source question. This is a bounded novelty check, not a claim that the construction is absent from all related literature.

Verification notes

The main proof to check is Theorem 1. Axiom (B2) is the only place where the Cotlar identity is used in a nontrivial way: from

$$m(b^{-1}a) \neq m(b^{-1}c)$$

one applies (1) with $g = c^{-1}b$ and $h = b^{-1}a$ to obtain

$$m(c^{-1}a) = m(c^{-1}b),$$

which says that c is not between a and b. Axiom (B3) is then just the elementary fact that if two colors differ, the color of a fourth point differs from at least one of them.

Human-review recommendation

Review as a likely valid partial result and as a promising route toward a full positive answer. The human verifier should focus on whether an existing pretree realization theorem can be quoted to turn this packet into a full Model 1 reverse construction.

References

[1] A. M. Gonzalez-Perez, J. Parcet, and R. Xia, *Noncommutative Cotlar identities for groups acting on tree-like structures*, arXiv:2209.05298.

[2] B. H. Bowditch, *Group actions on trees and dendrons*, Topology 37 (1998), no. 6, 1275–1298.

[3] A. V. Malyutin, *Pretrees and the shadow topology*, St. Petersburg Math. J. 26 (2015), no. 2, 225–271.

[4] E. Glasner and M. Megrelishvili, *Group actions on treelike compact spaces*, Science China Mathematics 62 (2019), 2447–2462; arXiv:1806.09876.